

Customer Churn Prediction & Revenue Forecasting

Using Supervised Machine Learning

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1. Executive Summary

This project builds a complete machine learning pipeline to predict customer churn for a telecom company and quantify the resulting revenue risk.

Using the IBM Telco Customer Churn dataset (7,032 customers, 21 features), three supervised learning models were trained and evaluated. Logistic Regression was selected as the final model based on its superior Recall score — the correct metric for this asymmetric cost problem where missing a churning customer is far more costly than a false alarm.

The revenue forecasting stage translates model predictions into dollar values, producing a prioritised list of at-risk customers and a retention campaign ROI model.

Key outcomes:

- Churn rate identified at 26.5% with clear high-risk customer segments
 - Model identifies the majority of churners before they leave
 - Monthly revenue at risk quantified with annual projection
 - Top 20 priority customers listed for immediate retention action
 - Break-even retention rate calculated for campaign planning
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2. Problem Statement

Business Context

Customer churn — when a customer stops using a service — is one of the most expensive problems in subscription-based businesses. Acquiring a new customer costs 5–7× more than retaining an existing one. For a telecom company, each churned customer represents lost monthly recurring revenue that is difficult and expensive to replace.

Objective

Build a model that predicts which customers are likely to churn, then quantify the revenue impact of those predictions to enable targeted, cost-effective retention campaigns.

Success Criteria

A model that catches as many actual churners as possible (high Recall) while providing accurate revenue risk estimates to prioritise retention spend.

3. Dataset Description

Property	Value
Source	IBM Telco Customer Churn (Kaggle)
Original size	7,043 rows × 21 columns
After cleaning	7,032 rows × 20 columns
Target variable	Churn (Yes/No → 1/0)
Churn rate	26.5% (1,869 churned, 5,163 retained)
Class imbalance	2.7:1 (retained:churned)

Feature categories:

- Customer demographics: gender, SeniorCitizen, Partner, Dependents
- Account details: tenure, Contract, PaperlessBilling, PaymentMethod
- Financial: MonthlyCharges, TotalCharges
- Services: PhoneService, InternetService, OnlineSecurity, and 6 more

4. Methodology — Data Science Pipeline

Stage 1: Data Exploration

Initial inspection revealed the dataset structure, data types, and the 26.5% churn rate. The class imbalance was documented and flagged for handling in model training.

Stage 2: Data Cleaning

Three cleaning operations were performed:

- `TotalCharges` column fixed from object to float64 dtype (hidden blank spaces)
- 11 rows with missing TotalCharges dropped (new customers, tenure=0)
- Binary categorical columns encoded to 0/1

Stage 3: Exploratory Data Analysis

Key findings (see figures 01–08 in reports/figures/):

- New customers (0–12 months) churn at ~47% vs 6% for long-tenure customers
- Month-to-month contracts have 43% churn vs 3% for two-year contracts
- Fiber optic internet customers churn at ~42% — double the average
- Churned customers pay ~\$13/month more than retained customers
- TotalCharges and tenure are highly correlated (0.83) — multicollinearity noted

Stage 4: Feature Engineering

- One-hot encoding applied to 7 multi-class categorical columns
- `drop_first=True` prevents the dummy variable trap

- 80/20 stratified train/test split (random_state=42)
- StandardScaler fit on training data only — no data leakage
- Final feature matrix: ~30 features

Stage 5: Model Training

Three models trained with 5-fold stratified cross-validation:

Model	CV Mean AUC	CV Std AUC
Logistic Regression	0.8459	0.0040
Random Forest	0.8479	0.0045
XGBoost	0.8381	0.0050

All models used `class_weight='balanced'` or equivalent to handle the 2.7:1 class imbalance.

Stage 6: Model Evaluation

Test set evaluation results (unseen data):

Model	ROC-AUC	Recall	Precision	F1
Logistic Regression	0.8353	0.7941	0.4901	0.6061
Random Forest	0.8376	0.7701	0.5363	0.6323
XGBoost	0.8206	0.7246	0.5192	0.6049

Model selection justification: Logistic Regression was selected despite not having the highest ROC-AUC. In churn prediction, a False Negative (missed churner) represents permanent revenue loss, while a False Positive (false alarm) results only in a small, recoverable retention cost. This asymmetric cost structure means Recall is the correct primary metric, and Logistic Regression achieves the highest Recall of the three models evaluated.

5. Revenue Forecasting Results

Core Formula

Revenue at Risk (per customer) = MonthlyCharges × Churn Probability

Summary

[Fill in from your Stage 7 Cell 5 output:]

- Total monthly revenue (test set): \$[amount]
- Expected revenue at risk: \$[amount]/month
- Annual projection: \$[amount]/year
- Percentage of revenue at risk: [X]%

Risk Tier Segmentation

Tier	Threshold	Customers	Revenue at Risk
High	>70%	378	\$24,153.86
Medium	40–70%	343	\$12,565.74
Low	<40%	686	\$6,082.97

Retention ROI

Targeting high-risk customers with a \$30/customer campaign:

- Break-even at approximately [X]% retention success rate
- At 50% retention success: \$[amount]/month net saved
- Annual net saving at 50%: \$[amount]/year

6. Business Recommendations

- Immediate (this week):** Contact the top 20 highest revenue-at-risk customers with personalised retention offers. These represent the highest ROI intervention available.
- Short term (this month):** Launch a structured retention campaign targeting all high-risk customers. Focus on converting month-to-month customers to annual contracts with an incentive offer.
- Medium term (this quarter):** Conduct a service quality audit for Fiber Optic infrastructure. The disproportionate churn rate among Fiber Optic customers suggests a product quality issue rather than a pricing problem.
- Ongoing:** Deploy the model as an automated monthly scoring system. Flag any customer crossing the 0.70 probability threshold for proactive outreach before they make the decision to leave.

7. Conclusion

- This project successfully demonstrates a complete, production-grade data science pipeline from raw data to business-actionable revenue forecasting.
- The selected Logistic Regression model provides the highest Recall among the three models evaluated, correctly prioritising the minimisation of missed churners over the minimisation of false alarms — the correct trade-off given the asymmetric costs in this business context.
- The revenue forecasting stage provides direct business value by converting probability scores into dollar figures, enabling cost-benefit analysis of retention campaigns and clear prioritisation of customer outreach.
- The high-risk segment contains only 26.9% of customers but contributes 56.4% of total revenue at risk, indicating strong risk concentration and making it the highest priority for retention efforts.

8. References

1. IBM Telco Customer Churn Dataset — Kaggle <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>
2. Scikit-learn Documentation — <https://scikit-learn.org>
3. XGBoost Documentation — <https://xgboost.readthedocs.io>
4. Chen, T. & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. KDD '16 Proceedings.